

ICCT MIS System Diagrams

Web-Based Resource Borrowing and Returning System

ICCT Cainta Campus — MIS Department

Revised: May 23, 2026

Figure 1 — System Architecture Diagram

The system follows a three-tier architecture supporting both web and mobile clients, with a Laravel 11 backend, MySQL database, Brevo email service, and cron-based task scheduling.

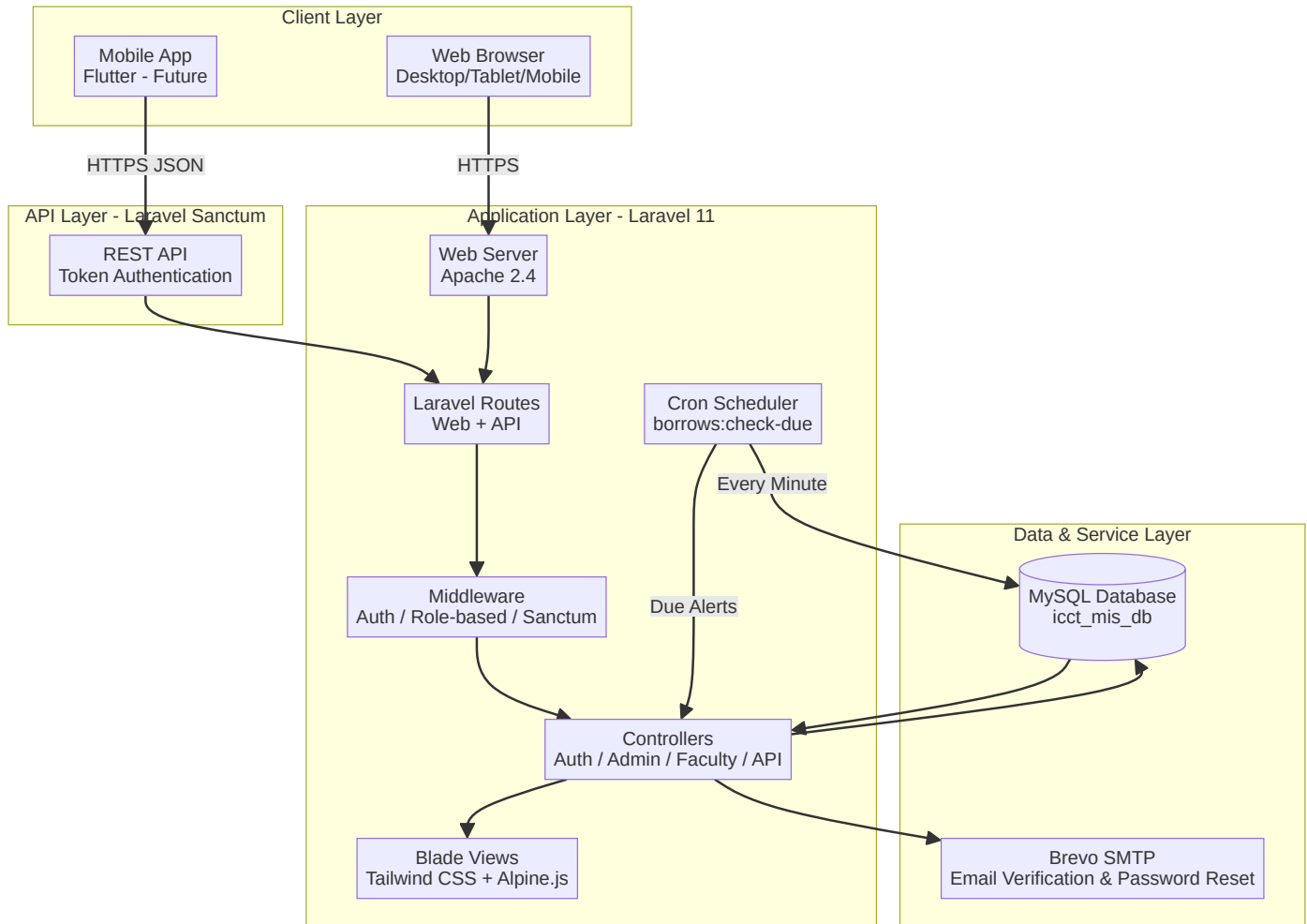


Figure 2 — Entity Relationship Diagram (ERD)

The database consists of 6 tables: users, resources, borrows, notifications, settings, and personal_access_tokens (Sanctum). The borrows table includes room and duration fields for tracking.

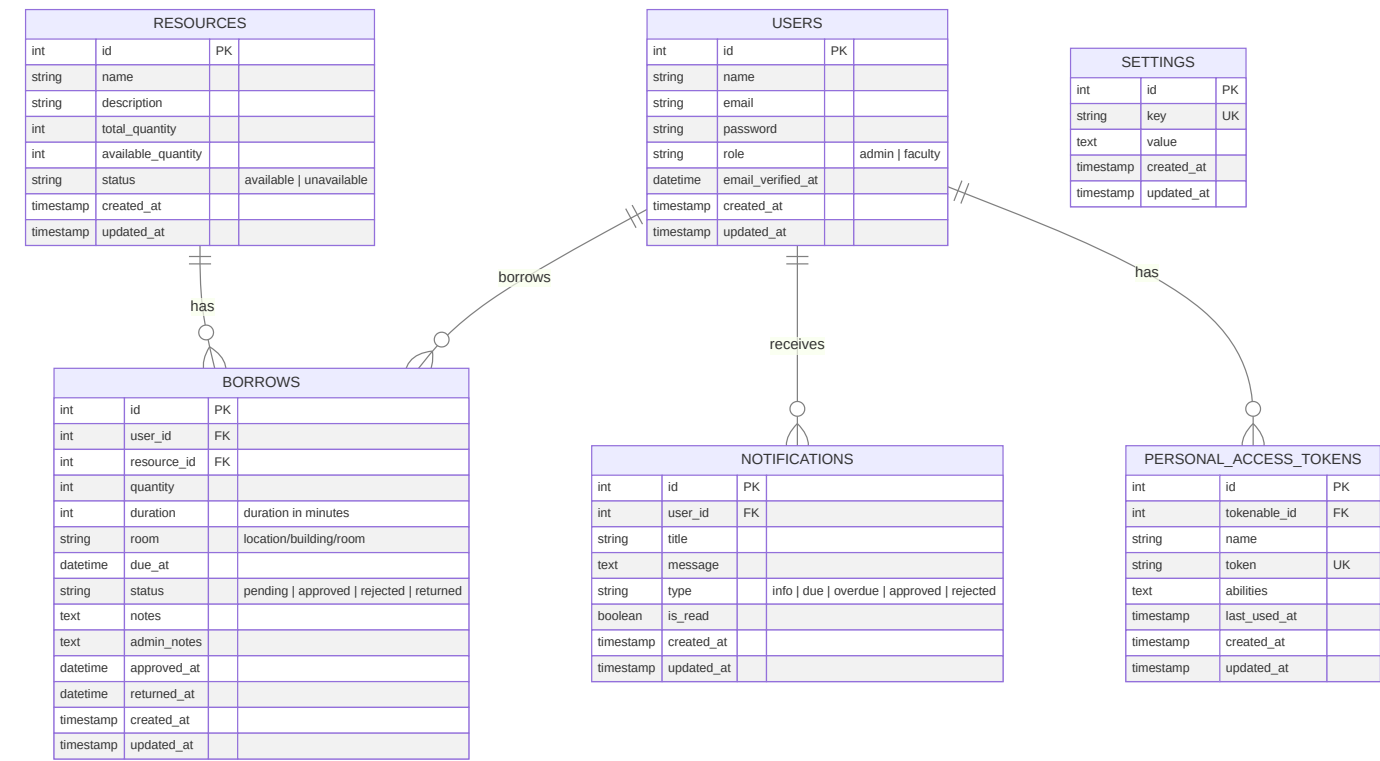
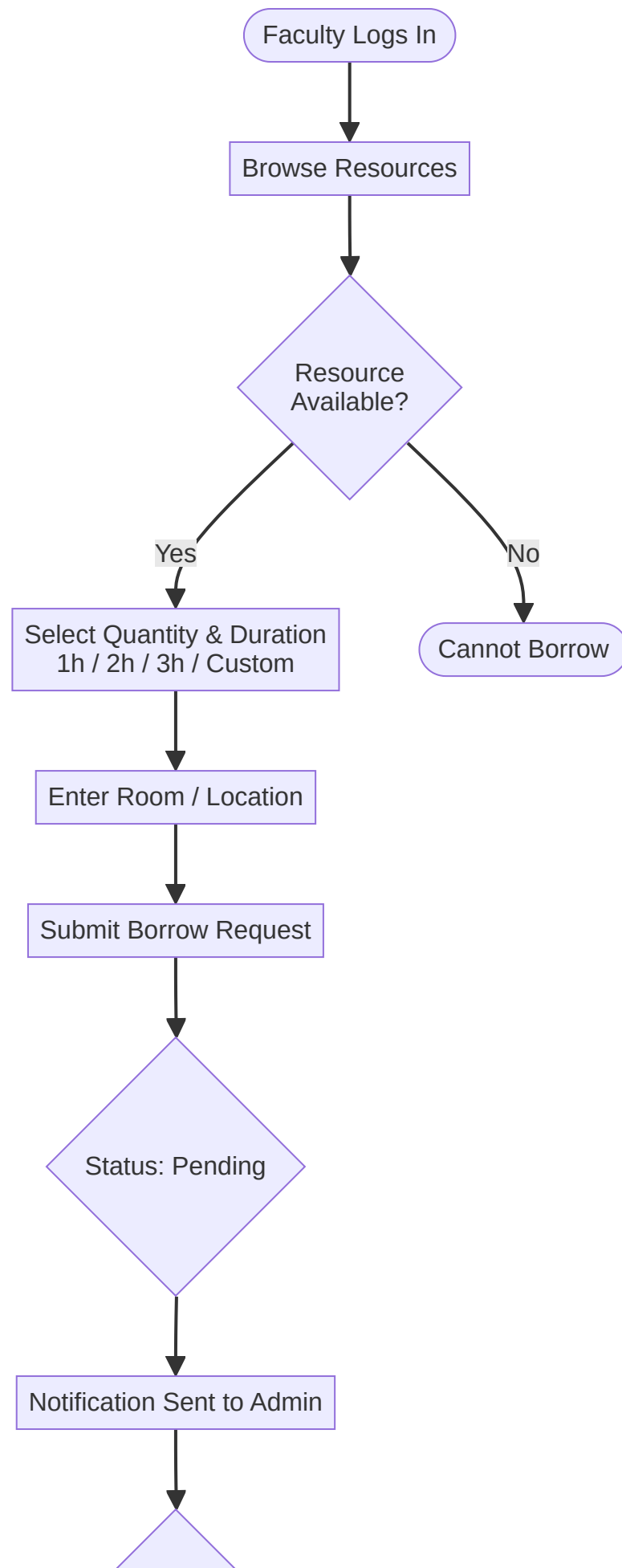
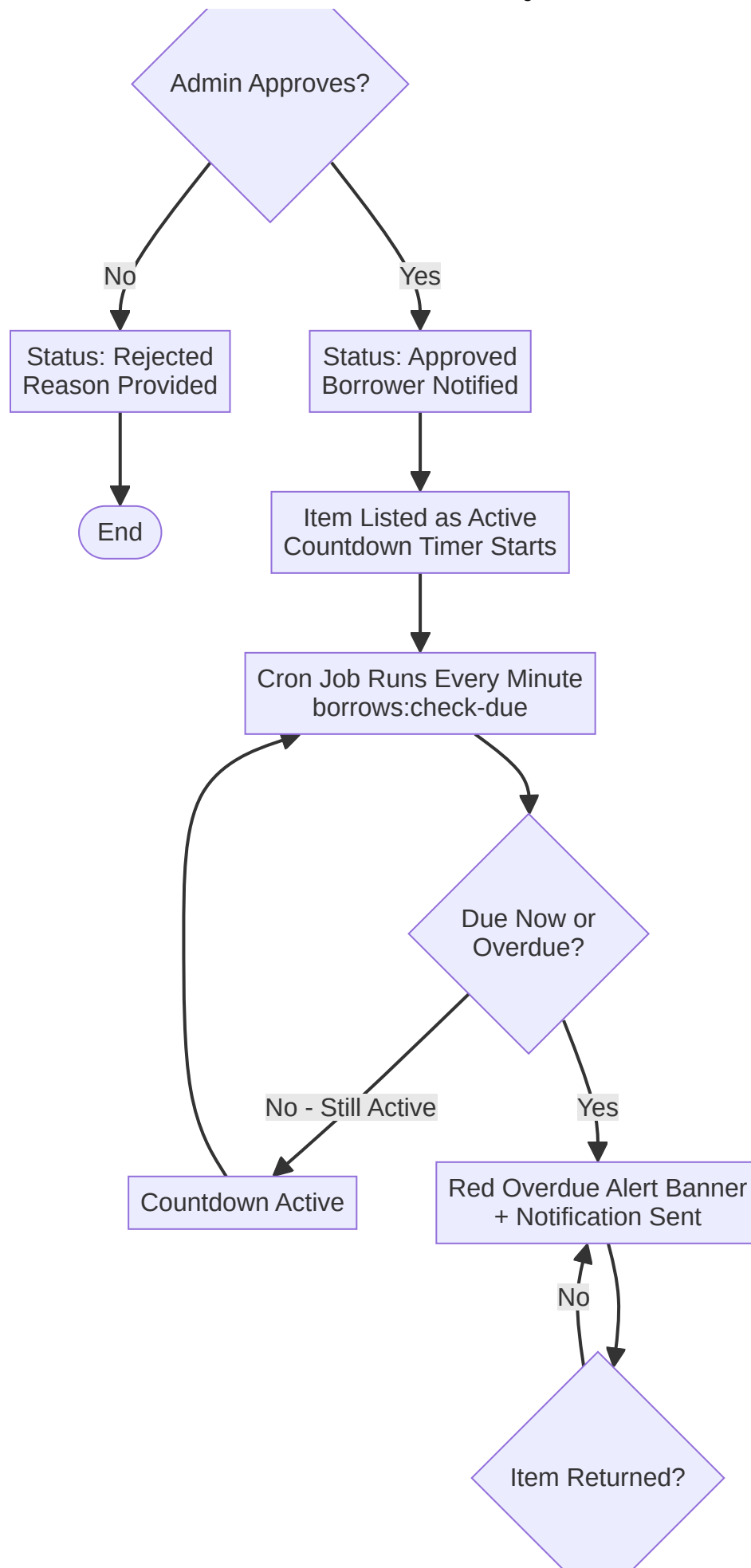


Figure 3 — Borrow Process Flowchart

The borrowing workflow shows the complete process from login through return. Includes room designation step and automated cron-based overdue detection.





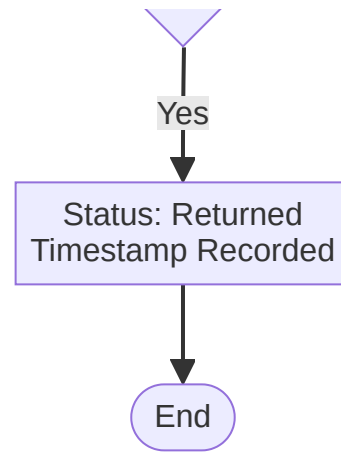
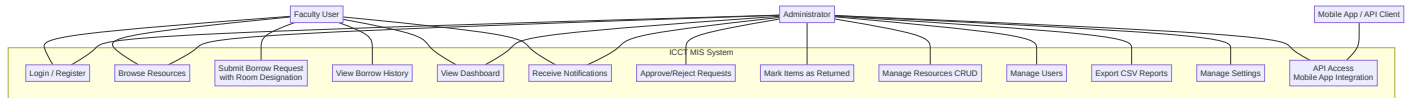


Figure 4 — Use Case Diagram

System functionality is divided between two user roles: Faculty and Administrator. Faculty can register, login, browse resources, submit borrow requests with room designation, and view history. Administrators have full management capabilities.



The data flow diagram shows how Faculty, Administrator, and Mobile App interact with system modules including the new API Gateway, Settings Manager, and Cron Monitoring modules.

